

Class: XI 'Science'

DAY SHIFT 'SET-A'

F.M: 37

Time: 1.5 hrs.

Subject: Physics [1011]

P.M: 15

Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempts all the questions.**GROUP - 'A' [6x1=6]****Multiple Choice Questions:**

- The correct formula for the displacement of a particle executing periodic motion is (v , T , a , y and t represent velocity, time period, amplitude, displacement and time respectively)
a) $y = a \sin \frac{2\pi}{T} t$ b) $y = a \sin vt$ c) $y = a \sin \frac{2\pi}{a} t$ d) $y = a \sin \frac{2\pi}{v} t$
- A vector remains unchanged on:
a) Rotating it through some angle
b) Taking its cross product with a unit vector
c) Taking its dot product with a unit vector
d) Shifting it parallel to itself
- When 2×10^{14} electrons are removed from a neutral metal sphere, the charge on the sphere becomes:
a) $-16 \mu\text{C}$ b) $16 \mu\text{C}$ c) $32 \mu\text{C}$ d) $-32 \mu\text{C}$
- A charge q_1 exerts some force on a second charge q_2 . If a third charge q_3 is brought near, the force q_1 exerted on q_2 :
a) Decreases b) Increases c) Remains unchanged
d) Increases if q_3 is of same sign as q_1 and decreases if q_3 is of the opposite sign.
- The dielectric constant for a metal is:
a) 1 b) 0 c) Greater than 1 d) infinity
- You are travelling in a car during a thunderstorm. In order to protect yourself from lightning would you prefer to:
a) Remain in the car b) Take shelter under a tree
c) Get out and lie flat on the ground d) Touch the nearest electric pole.

GROUP - 'B' [5X3=15]**Short Answer Questions:**

- a) Do all the physical quantities have dimensions? Justify your answer. [1]
b) In cgs system, the value of Stefan's constant σ is $5.67 \times 10^{-5} \text{ erg s}^{-1} \text{ cm}^{-2} \text{ K}^{-4}$. Find its value in international system of unit. [1]

- c) Check the correctness of the physical equations: i) $P = \frac{F}{A} + \rho gh$; P is pressure, F is force, A is area, ρ is density, g is acceleration due to gravity, and h, is height. ii) $T^2 = \frac{4\pi^2 a^3}{GM}$; T is time period, G is gravitational constant, M is mass and a is radius of the orbit. [3]
8. a) What is the necessary condition for adding of vectors. [1]
 b) Two forces act on a point and are represented by two sides of a triangle taken in order. How can you find the resultant force using this triangular representation? State and explain the law involved. [3]
 c) Under what condition is the magnitude of resultant force is zero Newton? [1]
9. a) Repulsion is the surest test of electrification. Explain? [2]
 b) What do you mean by electrostatic induction? How will you give permanent negative charge to a conductor by the method of induction? Explain in detail. [3]

GROUP - 'C' [2X8=16]

Long Answer Questions

10. a) A physical equation is dimensionally correct. Is it necessary that the equation is physically, correct? [1]
 b) Taking force (F), length (L), time (T), and current (A) as fundamental quantity, find the dimensional formula for permittivity ϵ_0 . [2]
 c) The gravitational constant is $G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$. Express G in terms of: Length in astronomical units ($1 \text{ AU} = 1.496 \times 10^{11} \text{ m}$), Mass in solar masses ($1 M_\odot = 1.989 \times 10^{30} \text{ kg}$), Time in years by dimensional method. [3]
 d) While deriving the relationship between physical quantities by dimensional analysis, dimensional constants enter into a relationship. Can you find its magnitude by the method of dimensions? If not, how will you find? [1]
 e) Mention any two limitations of the dimensional method. [1]
11. a) State and explain Coulomb's law of force between two stationary charges. Also, write the formula for Coulomb's force in vector form? What is the significance while expressing in vector form? [2+1+1]
 b) Two particles A and B having charges 4 nC and -1 nC respectively are held fixed with a separation of 0.20 m. Where should a third charged particle be placed so that it does not experience a net electrostatic force. [2]
 c) Two identical spheres having unequal opposite charges are placed at a distance 90 cm apart. After touching them mutually, they are again separated at the same distance apart. Now they repel each other with a force of 0.025 N. Calculate the final charge on each of them. [2]